

Geometry
Unit 3
Lesson 1 Homework

Name _____
Date _____
Period _____

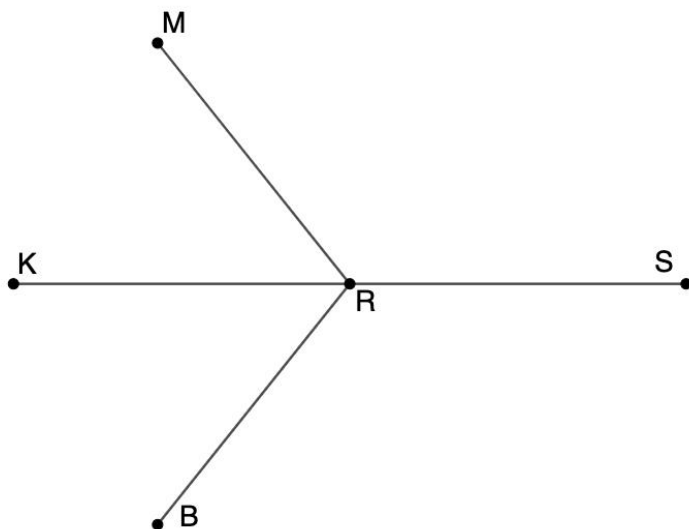
Use the phrases in the word bank to complete the proof. Some phrases may be used more than once.

Word Bank

Definition of Linear Pair	Linear Pair Postulate
Supplements of congruent angles are congruent.	Complements of congruent angles are congruent.
Angle addition postulate	Substitution property of equality
Definition of supplementary angles	Definition of complementary angles

Given: $\angle KRB \cong \angle KRM$

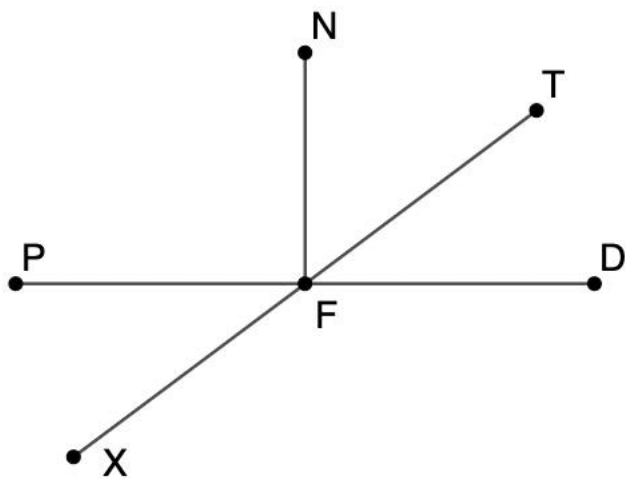
Prove: $\angle BRS \cong \angle MRS$



Complete the table using the word bank to prove $\angle BRS \cong \angle MRS$.

Statement	Reason
$\angle KRB \cong \angle KRM$	Given
$\angle KRB$ and $\angle SRB$ form a linear pair	1.
$\angle KRM$ and $\angle MRS$ form a linear pair	2.
$\angle KRB$ and $\angle SRB$ are supplementary angles	3.
$\angle KRM$ and $\angle MRS$ are supplementary angles	4.
$\angle BRS \cong \angle MRS$	5.

Line segments $\overline{PD}$, $\overline{NF}$, and $\overline{TX}$ intersect at point F as shown where $m\angle PFN = 90^\circ$, $m\angle XFP = (3x - 56)^\circ$, $m\angle NFT = (4z + 7)^\circ$ and $m\angle TFD = (x + 2)^\circ$.



6. Find the value of x .

7. Find the value of z .

8. What is $m\angle NFT$?

9. What is $m\angle XFD$?